File Guide

LinkedIn hiring loop — reverse-engineered

DOTE 6751 Database Systems · Group Project · November 2026

The map of the package: every file, how to rebuild, how to verify, and what to hand over.

1. The map

Generated by scanning the folder, so the list and the files agree.

```
DOTE6751_LinkedIn/
  File_Guide.docx   (42 KB)
  File_Guide.pdf   (199 KB)
  File_Guide_CN.docx   (43 KB)
  File_Guide_CN.pdf   (394 KB)
  README.md   (1 KB)
  Product_Spec/
    Product_Spec_CN.docx   (42 KB)
    Product_Spec_CN.md   (8 KB)
    Product_Spec_EN.docx   (41 KB)
    Product_Spec_EN.md   (8 KB)
  build/
    README.md   (2 KB)
    build_all.py   (2 KB)
    build_deck.py   (44 KB)
    build_file_guide.py   (23 KB)
    build_narrative_docx.py   (15 KB)
    build_queries.py   (4 KB)
    erd.py   (15 KB)
    fault_injection.py   (5 KB)
    gen_normalization_xlsx.py   (24 KB)
    gen_seed.py   (42 KB)
    queries_template.sql   (23 KB)
    run_queries.py   (2 KB)
    verify_data.py   (8 KB)
  design/
    Constraint_Inventory.md   (10 KB)
    Design_Rationale.md   (12 KB)
    Entity_Relationship_Catalog.md   (13 KB)
    Normalization_Proof.xlsx   (29 KB)
    Relational_Schema.md   (11 KB)
    Requirements.md   (8 KB)
  erd/
    ERD_conceptual.png   (359 KB)
    ERD_conceptual.svg   (83 KB)
    ERD_logical_full.png   (267 KB)
    ERD_logical_full.svg   (179 KB)
    ERD_module_demand.png   (135 KB)
    ERD_module_demand.svg   (92 KB)
    ERD_module_hiring_loop.png   (152 KB)
    ERD_module_hiring_loop.svg   (103 KB)
    ERD_module_identity.png   (124 KB)
    ERD_module_identity.svg   (97 KB)
  query_results/
    _consistency_report.json   (3 KB)
    _summary.json   (6 KB)
```

```
  q01_result.csv   (<1 KB)
  q01_result.png   (84 KB)
  q02_result.csv   (<1 KB)
  q02_result.png   (100 KB)
  q03_result.csv   (2 KB)
  q04_result.csv   (<1 KB)
  q05_result.csv   (<1 KB)
  q05_result.png   (70 KB)
  q06_result.csv   (<1 KB)
  q07_result.csv   (<1 KB)
  q08_result.csv   (<1 KB)
  q09_result.csv   (<1 KB)
  q09_result.png   (76 KB)
  q10_result.csv   (<1 KB)
  q11_result.csv   (<1 KB)
  q12_result.csv   (<1 KB)
  q12_result.png   (73 KB)
  q13_result.csv   (<1 KB)
  q14_result.csv   (<1 KB)
  q15_result.csv   (1 KB)
  q16_result.csv   (<1 KB)
  q17_result.csv   (1 KB)
  q18_result.csv   (<1 KB)
  q18_result.png   (75 KB)
slides/
  LinkedIn.pptx   (1001 KB)
  LinkedIn_CN.pptx   (1003 KB)
  LinkedIn_Narrative.docx   (41 KB)
  LinkedIn_Narrative.pdf   (95 KB)
  Presentation_Script.md   (10 KB)
  QA_Prep.md   (8 KB)
  Task_Allocation.md   (7 KB)
sql/
  00_run_all.sql   (<1 KB)
  01_schema.sql   (17 KB)
  02_seed.sql   (298 KB)
  03_queries.sql   (23 KB)
  04_views_triggers.sql   (10 KB)
  05_transactions.sql   (4 KB)
  06_checks.sql   (5 KB)
  linkedin_db_full.sql   (274 KB)
```

2. Every file, and what it is for

Grouped by folder. What matters is when you would open it.

Root

File	Size	What it is	What is inside
File_Guide.docx	42 KB	—	—
File_Guide.pdf	199 KB	—	—
File_Guide_CN.docx	43 KB	—	—
File_Guide_CN.pdf	394 KB	—	—
README.md	1 KB	Quick orientation	Three sentences on what the package is and where to start.

Product_Spec/

File	Size	What it is	What is inside
Product_Spec_CN.docx	42 KB	Product spec (CN)	Positioning, vision, substitutes, scenarios, features, economics, metrics, risks, roadmap.
Product_Spec_CN.md	8 KB	Product spec source (CN)	The same document as Markdown.
Product_Spec_EN.docx	41 KB	Product spec (EN)	The same document in English.
Product_Spec_EN.md	8 KB	Product spec source (EN)	The same document as Markdown.

build/

File	Size	What it is	What is inside
README.md	2 KB	Quick orientation	Three sentences on what the package is and where to start.
build_all.py	2 KB	Build script	Regenerates the artefact; not a deliverable.
build_deck.py	44 KB	Build script	Regenerates the artefact; not a deliverable.
build_file_guide.py	23 KB	Build script	Regenerates the artefact; not a deliverable.
build_narrative_docx.py	15 KB	Build script	Regenerates the artefact; not a deliverable.
build_queries.py	4 KB	Build script	Regenerates the artefact; not a deliverable.
erd.py	15 KB	Build script	Regenerates the artefact; not a deliverable.
fault_injection.py	5 KB	Build script	Regenerates the artefact; not a deliverable.
gen_normalization_xlsx.py	24 KB	Build script	Regenerates the artefact; not a deliverable.
gen_seed.py	42 KB	Build script	Regenerates the artefact; not a deliverable.
queries_template.sql	23 KB	Build script	Regenerates the artefact; not a deliverable.
run_queries.py	2 KB	Build script	Regenerates the artefact; not a deliverable.
verify_data.py	8 KB	Build script	Regenerates the artefact; not a deliverable.

design/

File	Size	What it is	What is inside
Constraint_Inventory.md	10 KB	Constraint inventory	Every rule, its type, where it is enforced, and what happens when it breaks.
Design_Rationale.md	12 KB	Design rationale	Why these tables, why the weak entity, why derived values are stored, what was rejected.
Entity_Relationship_Catalog.md	13 KB	E-R catalogue	Every entity and relationship written out: cardinality, participation, attributes.
Normalization_Proof.xlsx	29 KB	Normalization proof	Functional dependencies, candidate keys and the 3NF/BCNF argument, per table.
Relational_Schema.md	11 KB	Relational schema	The E-R design translated into relations, with keys and mapping rules.
Requirements.md	8 KB	Requirements and scope	The problem, the actors, the functional requirements, scope

			and assumptions.

erd/

File	Size	What it is	What is inside
ERD_conceptual.png	359 KB	Conceptual E-R	Entities, relationship diamonds and cardinality.
ERD_conceptual.svg	83 KB	Vector version	Scales without blur; opens in a browser.
ERD_logical_full.png	267 KB	Logical schema	Every table with columns, keys and connectors.
ERD_logical_full.svg	179 KB	Vector version	Scales without blur; opens in a browser.
ERD_module_demand.png	135 KB	Module E-R	A zoomed view of that module.
ERD_module_demand.svg	92 KB	Vector version	Scales without blur; opens in a browser.
ERD_module_hiring_loop.png	152 KB	Module E-R	A zoomed view of that module.
ERD_module_hiring_loop.svg	103 KB	Vector version	Scales without blur; opens in a browser.
ERD_module_identity.png	124 KB	Module E-R	A zoomed view of that module.
ERD_module_identity.svg	97 KB	Vector version	Scales without blur; opens in a browser.

query_results/

File	Size	What it is	What is inside
_consistency_report.json	3 KB	Consistency report	Results of the cross-table invariants.
_summary.json	6 KB	Query summary	Rows, timing and columns per query.
q01_result.csv	<1 KB	Query 1 result	The full output of that query on the seeded instance.
q01_result.png	84 KB	Query 1 result image	The result rendered as a table image.
q02_result.csv	<1 KB	Query 2 result	The full output of that query on the seeded instance.
q02_result.png	100 KB	Query 2 result image	The result rendered as a table image.
q03_result.csv	2 KB	Query 3 result	The full output of that query on the seeded instance.
q04_result.csv	<1 KB	Query 4 result	The full output of that query on the seeded instance.
q05_result.csv	<1 KB	Query 5 result	The full output of that query on the seeded instance.
q05_result.png	70 KB	Query 5 result image	The result rendered as a table image.
q06_result.csv	<1 KB	Query 6 result	The full output of that query on the seeded instance.
q07_result.csv	<1 KB	Query 7 result	The full output of that query on the seeded instance.
q08_result.csv	<1 KB	Query 8 result	The full output of that query on the seeded instance.
q09_result.csv	<1 KB	Query 9 result	The full output of that query on the seeded instance.
q09_result.png	76 KB	Query 9 result image	The result rendered as a table image.
q10_result.csv	<1 KB	Query 10 result	The full output of that query on the seeded instance.
q11_result.csv	<1 KB	Query 11 result	The full output of that query on the seeded instance.
q12_result.csv	<1 KB	Query 12 result	The full output of that query on the seeded instance.

q12_result.png	73 KB	Query 12 result image	The result rendered as a table image.
q13_result.csv	<1 KB	Query 13 result	The full output of that query on the seeded instance.
q14_result.csv	<1 KB	Query 14 result	The full output of that query on the seeded instance.
q15_result.csv	1 KB	Query 15 result	The full output of that query on the seeded instance.
q16_result.csv	<1 KB	Query 16 result	The full output of that query on the seeded instance.
q17_result.csv	1 KB	Query 17 result	The full output of that query on the seeded instance.
q18_result.csv	<1 KB	Query 18 result	The full output of that query on the seeded instance.
q18_result.png	75 KB	Query 18 result image	The result rendered as a table image.

slides/

File	Size	What it is	What is inside
LinkedIn.pptx	1001 KB	Deck (EN)	Fourteen slides with speaker notes.
LinkedIn_CN.pptx	1003 KB	Deck (CN)	The same deck in Chinese.
LinkedIn_Narrative.docx	41 KB	Narrative	About 1,600 words across six sections.
LinkedIn_Narrative.pdf	95 KB	Narrative PDF	The same text.
Presentation_Script.md	10 KB	Speaking script	What to say on each slide, with timings and speaker split.
QA_Prep.md	8 KB	Q&A preparation	Likely questions with answers short enough to say aloud.
Task_Allocation.md	7 KB	Task allocation	Seven technical tasks with difficulty and hours; no presentation role.

sql/

File	Size	What it is	What is inside
00_run_all.sql	<1 KB	Rebuild script	Calls the other scripts in the only correct order.
01_schema.sql	17 KB	DDL	Creates the database and every table with keys, checks and indexes.
02_seed.sql	298 KB	Seed data	Deterministic data; ends with a reconciliation statement.
03_queries.sql	23 KB	Eighteen queries	Three families; each statement runs on its own.
04_views_triggers.sql	10 KB	Views and triggers	Read models and the triggers that enforce business rules.
05_transactions.sql	4 KB	Transaction episodes	Self-contained episodes, rolled back, so the data is unchanged.
06_checks.sql	5 KB	Acceptance checks	Twelve checks; every row must read PASS.
linkedin_db_full.sql	274 KB	Full dump	Schema, data, views and triggers in one file.

3. How to rebuild the database

Navicat: Run SQL File → sql/linkedin_db_full.sql.

Or in order: 01_schema → 02_seed → 04_views_triggers → 06_checks → 03_queries.

Command line: mysql -u root -p < 00_run_all.sql

Order matters: the seed contains applications withdrawn after interviews, which the guards would refuse.

4. How to verify

Eight steps; the first six take three minutes.

#	Check	How	A pass looks like
1	Service running	MySQL80 Running	Started
2	Client connects	Test Connection	Success
3	Schema complete	Restore and expand	12 tables, 4 views, 4 triggers
4	Data arrived	Count members / applications / interviews	300 / 250 / 210
5	Acceptance script	Run 06_checks.sql	12 rows, all PASS
6	Queries answer	Run 03_queries.sql	q01 seven rows, q09 twelve
7	Constraints refuse	Run the three below	Each fails
8	Invariants hold	Open _consistency_report.json	22 of 22 pass

These three are meant to fail. If one succeeds, a constraint is missing.

Referral from outside

```
INSERT INTO referral (application_id, referrer_id, relationship, referred_at, bonus_amount,
bonus_status)
VALUES (1, 1, 'OTHER', NOW(), 0, 'NONE');
```

Expected: error 1644, referrer not employed

Interview too early

```
INSERT INTO interview (application_id, round_no, interview_type, scheduled_at, duration_min,
interviewer_id, outcome)
VALUES (9, 1, 'PHONE', NOW() + INTERVAL 2 DAY, 30, 1, 'PENDING');
```

Expected: error 1644, stage not reached

Duplicate application

```
INSERT INTO application (job_id, member_id, source_channel, status, applied_at)
VALUES (1, 1, 'DIRECT', 'SUBMITTED', NOW());
```

Expected: error 1062, duplicate key

5. What was checked before delivery

Twelve checks, all executed.

#	Check	Violations	Result
C01	Twelve tables present	12	PASS

C02	Four views and four triggers	8	PASS
C03	No decision before application	0	PASS
C04	Terminal rows have a date	0	PASS
C05	In-flight rows have none	0	PASS
C06	Hires within openings	0	PASS
C07	No self-connection	0	PASS
C08	No duplicate connections	0	PASS
C09	Proficiency in 1–5	0	PASS
C10	Scores in 1–5	0	PASS
C11	Referrals from inside	0	PASS
C12	One current employer each	0	PASS

6. Decisions worth knowing

Twelve tables

Eleven entity sets plus the self-referential connection; the current employer is not a column.

The current employer is derived

experience.is_current plus a unique index on a generated column.

Referral is 1:1 with its application

It stores only the referrer and the bonus.

Guards live in the database

Referrer employment and interview stage cannot be bypassed by any client.

The data is synthetic

Plausible distributions, not empirical; every finding describes the seeded instance.